EVENT RECOMMENDATION SYSTEM

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ABSTRACT

This project presents the hybrid model of an event recommendation system using machine learning and data mining techniques. Leveraging collaborative and content-based filtering methods, it offers personalized recommendations by analysing user interests and event characteristics. Collaborative filtering identifies similar users and recommends events enjoyed by others with similar interests, while content-based filtering focuses on event attributes to make relevant suggestions. By combining these approaches, the system provides diverse recommendations. The dataset used in the project comprises 'events.csv' and 'train.csv', containing event details and user-event interests, respectively. The preprocessing step in the project involves handling missing values and feature engineering, resulting in datasets ready for modelling. The model of the event recommendation system consists of two sub-models:

Model 1: Identifies similar users based on their responses to events, employing Nearest Neighbours.

Model 2: Focuses on finding similar events using cosine similarity.

The final model combines outputs from both models, generating event recommendations personalized to user preferences.

Keywords: Collaborative Filtering, Content-Based Filtering, Data preprocessing, Cosine Similarity, K-Nearest Neighbour.

1. INTRODUCTION

In today's fast-paced world, there are countless events happening, covering a wide range of interests and locations. Whether it's cultural festivals, sports events, professional conferences, or community gatherings, which provide engaging experiences. But with so many events available, it can be overwhelming to find the right events to attend. That's where event recommendation systems come in. They're like personalized guides that help people discover events that match the user’s interests.

Today, many event recommendation systems fall short of delivering diverse personalized suggestions. One common issue is that some systems rely solely on basic demographic information or past event attendance, failing to capture the interests of users. Some recommendation systems use rudimentary methods like popularity-based or random recommendations, which may not reflect users' individual interests or provide relevant suggestions.

Furthermore, the lack of context-awareness is a common limitation. Ineffective event recommendation systems often suffer from poor user engagement and feedback mechanisms. Without robust mechanisms for collecting and incorporating user feedback (such as them being “interested” or “not interested” in the event), these systems struggle to learn and adapt over time, resulting in stagnant and unimproved recommendations.

Overall, the ineffectiveness of many event recommendation systems stems from lack of context-awareness. Addressing this drawback is essential for developing more robust and personalized event recommendation solutions in the future.

The Event Recommendation System we're discussing here is a hybrid solution to this challenge. It uses advanced techniques of machine learning and data mining, to give users relevant suggestions for events they might be interested. Fundamentally, this hybrid system model combines two methods: collaborative filtering and content-based filtering methods. Collaborative filtering looks at what similar users are interested in to recommend events, while content-based filtering focuses on the details of each event to make suggestions based on their characteristics.

To make all this happen, we use detailed datasets with information about events, user interactions, and preferences. We carefully clean and organize this data so that the system can understand it better and give accurate recommendations. The system has two main parts: Model 1 finds users with similar interests and suggests events they might like, and Model 2 identifies events that are similar to each other and recommends them based on their descriptions.

2. LITERATURE REVIEW

Recommender systems often face challenges with sparse data and accuracy. To address this issue, a hybrid approach using both collaborative filtering (CF) and content-based (CB) methods was proposed by Cornelis Et al. [1]. In this study, fuzzy relations were used to understand how users pick the events. This method identifies detailed preferences and calculates similarity by considering both item features and user actions. Results show that this approach is more effective than the existing methods in providing accurate recommendations tailored to individual user needs. In a later study by Khrouf and Troncy [5] suggested that combining content-based (CB) and collaborative filtering (CF) methods and using Linked Data to recommend events and statistical user interest model to handle the diversity of user preferences. The recommendations were made using a hybrid model which combines both Content-Based and collaborative filtering. The evaluation showed improvement in precision and recall with Linked Data usage and user diversity modelling. Collaborative filtering notably improved recommendation accuracy. The results suggest that the hybrid algorithm performs better than traditional Collaborative Filtering and extended profile filtering methods. Pessemier Et al. [4] used the Outlife recommender algorithm to streamline this process by categorizing Facebook friends into clusters based on interaction patterns and proposing personalized event suggestions. Hierarchical clustering was used for identifying friend groups and the user's closest connections through Facebook likes and tags. Results revealed that 64% of users found the model to be accurate in suggesting friends, with 71% subsequently attending events with the recommended individuals. In another study, Klamma Et al. [2] have suggested a solution by introducing a model that pulls together information from digital libraries and the internet to help recommend events and analyze the groups with common interests around them. The model tries to identify both the events and the user groups that could be interested in the event by using Actor-Network Theory and Social Network Analysis. It then applies collaborative filtering to suggest events based on a user's history. The results showed that most found the recommendations and community insights provided by the system were helpful. Daly and Geyer [3] aimed to improve event recommendation accuracy by using both online and physical interactions by sorting events based on the user's location and social aspects, using data from an organization's event management service. K-Nearest Neighbor based collaborative filtering approach with the Log-likelihood Similarity model was used to study the user attendance patterns and to suggest events based on similar user preferences. The authors have also used methods for event classification based on social and location factors. Social scoping looks at past attendee overlap, while location scoping is calculated using event locations from attendee geographic data. These approaches were used to better understand event characteristics and improve recommendation accuracy. The analysis reveals a strong representation of events with high social interaction but without a local focus on recommendations. Including location information helps make recommendations more accurate, which is especially helpful for people with no history. In another study by Macedo Et al. [6], the study was particularly focused on cold-start events where user data is limited. This paper uses user-group relations, content-based filtering, location preferences, and temporal patterns to recommend events. The authors have used Multi-Relational Factorization with Bayesian Personalized Ranking (MRBPR) to rank. The results suggest that using diverse contextual factors for event recommendations is effective and improves the recommendation accuracy. In a later study by Boutsis Et al [7], the Mixed Markov Model was used to extract the behavioral patterns of the users in social groups, to make personalized event recommendations. This study also addressed challenges such as cold start issues and sequential event attendance. The model used group behaviour from social network data to predict user event attendance and used features like attendance frequency and location centroids to divide users into groups for accurate predictions. The model is evaluated using collaborative filtering and historical visits approaches. The model had an event prediction accuracy of up to 85.8% over other techniques and this model could also effectively address the cold start problem by using group behavior which shows practicality and efficiency with real-world data. In another study by Ogundele Et al. [8], the authors tried to increase the accuracy of event recommendations in social networks by considering geographical, social, and temporal influences. Kernel Density Estimation (KDE) was used to find out the event location influence on users, have used the Jensen-Shanon divergence to measure similarity between event times and past user activities, Collaborative Filtering for social influence, and Singular Value Decomposition (SVD) for event attendance prediction. This entire set of models is named as Eventrec, and it outperformed existing methods across precision and recall metrics. The EventRec model has shown a precision of 0.75 and a recall of 0.68 when compared to other methods. In a later study by Bok Et al. [9], the authors addressed the inefficiency of existing event recommendation systems in social network environments, particularly when recommending events to users with limited event participation history. This event recommendation method uses a combination of techniques which includes collaborative filtering, matrix factorization (MF), and feature value prediction, to generate event recommendations for individual users. The authors have analysed user participation history to predict preferences, using Matrix Factorization to estimate unevaluated features and then employ collaborative filtering to generate personalized event sets. This method has outperformed the existing models by 10-30%, user satisfaction scores were 10% higher, and precision increased by over 30%. The F-Measure improved by approximately 20%, which indicates higher accuracy and satisfaction.

In this project, we utilized both Cosine Similarity and K-Nearest Neighbors to improve accuracy and address sparse data issues.

3. METHODOLOGY

The main components of this project are

Data Visualization:

Data visualization plays a crucial role in understanding the underlying patterns and relationships within the dataset. In this project, data visualization serves in identifying the % of missing data and will that be useful if followed a way to replace the missing values. And helps in identifying which are predictive, interative, redundant and irrelevant features.

Data Preprocessing:

Bases on the information about the features we dropped few features and prepared the data for the task.

Model 1:

Model 1 is to identify the similar users bases on the response ‘interested or not interested' for an event. This is achieved using the K-Nearest Neighbors algorithm, a non-parametric method that identifies users with response patterns most similar to the target user. By leveraging the KNN algorithm, Model 1 effectively captures the local structure of the data, allowing for personalized recommendations based on user preferences.

Error Analysis and parameter tuning:

Sine KNN is prone to overfitting and underfitting. RMSE is used to calculate the error value and did parameter tuning for finding best n-neighbour value using 10-fold Cross-validation.

Model 2:

Model 2 is to find similar events. This is achieved by using Cosine similarity. Cosine similarity measures how similar two vectors are in direction, regardless of their magnitude. It's useful for comparing text data and high-dimensional data. It is also computationally efficient, especially for sparse data. Sine the dataset we are dealing in this project is having different magnitude and high-dimensional spaces.

With the help of cosine similarity, similarity matrix is obtained for events.

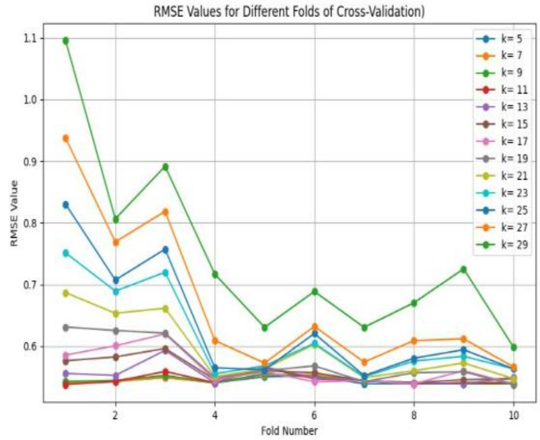
Final Mode:

Final model is combining both model 1 and model 2 outputs. The statuary is to obtain the list of events that is interesting for a user and his similar users. For those events similar events with similarity greater than 0.5 is obtained. And these events are sorted based on the similarity score and top 100 events were suggested for the users.

4. IMPLEMENTATION

The project required us to use 2 data sets from a challenge on Kaggle. One called “events” and it is large events dataset which contained 110 features: 'event_id', 'user_id', 'start_time', 'city', 'state', 'zip', 'country', 'lat', 'lng', 'c_1',… 'c_98', 'c_99', 'c_100', 'c_other'. The c-values are a word count in

the description of the event. However, as part of data preprocessing we need to drop city, state, zip, country, lat, lng, user_id, and start_time as they were missing over half of their values reaplacing missing with other values don’t help here because most of them are location details and other few are the details on the user who created the event that don't add any value in our task and the other dataset is “train” with 6 features: 'user', 'event', 'interested', 'not_interested', 'timestamp', 'invited' which housed each user's opinion on events. The KNN algorithm was used on the train file to find the closely related users. During the implementation of the KNN algorithm, we conducted error analysis to find the best value of “k”. Finding the appropriate “k” value allows us to obtain the most relevant results. To carry out the error analysis, we made use of the RMSE (Root Mean Square Error) curve, plotting for each fold under each cross validation against its RMSE value. As part of the analysis, we have used the fine-tuned KNN instead of the normal KNN. This is because fine-tuning KNN for error analysis offers benefits such as improved performance through optimizing hyperparameters like the number of neighbors. This optimization enhances the model's ability to generalize to new data while reducing overfitting by finding the right balance between bias and variance. Moreover, fine-tuning provides insights into the data patterns, making the model more interpretable. Additionally, it allows for customization tailored to specific problem domains, offering flexibility in achieving desired performance levels. In essence, fine-tuning KNN enables more effective error analysis and decision-making. Shown below is the RMSE graph obtained from the analysis.
After analyzing the RMSE graph, we find that when the “k” value is 11, we get the least RMSE errors. Thus, we set the “k” value to 11 in the KNN algorithm. Using this, closely related users were found.



RMSE graph for the various K values at each fold against the RMSE values.

This also prevents the model from underfitting or overfitting issues. Later in model-2, cosine similarity was used on the events dataset which found which events were similar. Then after finding an event that each user liked, cosine similarity matrix was used to find the closest related events to that event giving each user 100 recommendations.
However, after some review the similarities between users because of their lack of responses.

event	user	1000481836	1001013482	1002636124	1002911800	1005830738	1006903887	1009755933
0	1000293064	0	0	0	0	0	0	0
1	1006838695	0	0	0	0	0	0	0
2	1008893291	0	0	0	0	0	0	0
3	1011223331	0	0	0	0	0	0	0
4	1016040879	0	0	0	0	0	0	0

5 rows × 3207 columns

The above is the sparse dataset of User-event response

So, to make the project more accurate this was cut. This left us with an event similarity model which just grouped the similar events based off cosine similarity. This model would give all the similar events in order from highest similarity score at first to the lowest at last. And for each user bases on the event, they responded as interested into consideration and similar for that event is grabbed and given as suggestion for each user.

5. RESULTS AND DISCUSSION

The Hybrid model recommendation where less accurate because of the sparse data in user's dataset. But our model tried it best to recommendation even for each user. But this recommendation is not fully accurate.

user	0	1	2	3	4	5
1000293064	3900421971	3755008121	97217712	2027514460	3635358150
1006838695	3338163812	3755008121	97217712	2027514460	3900421971
1008893291	2307226742	3921231623	1637337509	830994161	3460795809
1011223331	2307226742	3921231623	1637337509	830994161	3460795809
1016040879	2307226742	3921231623	1637337509	1144728004	1100725936

user	6	7	8	9	...	90
1000293064	2285108120	764214719	4002262960	3425996117	...	3439956212
1006838695	2307226742	1619986209	1637337509	2452904679	...	3166241555
1008893291	1436270591	1221934300	3062540233	44776435	...	843570927
1011223331	1436270591	1221934300	3062540233	44776435	...	4021236305
1016040879	830994161	3460795809	1436270591	1221934300	...	3191533412

user	91	92	93	94	95
1000293064	3950286482	963138294	449812514	1555919109	3974620202
1006838695	4210789941	2583924875	239276659	3482133976	2622371373
1008893291	3782727355	2863056708	3446902030	4021236305	702157839
1011223331	702157839	3080864770	1281417389	3328980614	811345080
1016040879	546889678	1095332232	1769487166	1061572648	843570927

user	96	97	98	99
1000293064	4044222914	4235989936	1306613841	141144704
1006838695	3648276654	298169907	1929622843	2499367081
1008893291	3080864770	1281417389	3328980614	1366554501
1011223331	3338163812	3755008121	1619986209	2351245308
1016040879	3782727355	835360674	2863056708	3446902030

[5 rows x 100 columns]

Above is the recommendation for each user using Hybrid model.

Alternate Model:

Since the hybrid model's recommendations are not highly accurate, we tried to produce a model that gives event recommendations for user bases on event they responded as interested.

The Event Recommendation System yielded promising results in enhancing event recommendation effectiveness. By leveraging advanced machine learning techniques and data preprocessing strategies, the system successfully provided event recommendations. This is the recommendation for each event. The results will have more relevant event in the first index and less relevant event in the last index.

	Event	Similar_Event_1	Similar_Event_2	Similar_Event_3	Similar_Event_4
0	684921758	585272047	3438993032	845902488	4192218834
1	244999119	2604545919	1018819627	2700038507	590379964
2	3928440935	2005050320	472699259	1095858231	1455527953
3	2582345152	40348900	3689283674	380501274	3585709696
4	1051165850	3028497756	1885776015	2876844499	297856962

	Similar_Event_5	Similar_Event_6	Similar_Event_7	Similar_Event_8
0	1973916091	1781986615	1091130052	2233735163
1	1076499062	1769821149	2401113510	3161434996
2	546889678	1364538695	4163453351	3703612480
3	3917524398	3050501263	3701432790	2918749132
4	2998372996	895832538	1333829432	3882632768

	Similar_Event_9	...	Similar_Event_91	Similar_Event_92	Similar_Event_93
0	3596510482	...	2602203758	898889852	1413634049
1	690062686	...	4133786107	1343362100	920467258
2	3716036142	...	2396566946	3039059830	2588877820
3	4020625664	...	2821117540	1907550340	31247346
4	2622371373	...	4219761316	3472605389	2573061069

	Similar_Event_94	Similar_Event_95	Similar_Event_96	Similar_Event_97
0	1859823732	4247882282	1983888102	1396446223
1	2165261948	4092423355	2437040715	1769487166
2	2144335101	892306356	3145208730	3331152144
3	3744389392	3755008121	2575303578	3162166853
4	1091130052	2624001814	1066839399	3773489407

	Similar_Event_98	Similar_Event_99	Similar_Event_100
0	42204521	1381535648	3585709696
1	2398136361	2871310940	3580637647
2	944759264	3585709696	3060412046
3	3039059830	2644800408	2700038507
4	1095858231	31247346	550528378

[5 rows x 101 columns]

Connected to Python 3 Google Compute Engine backend

The above shows the events recommended.

Above is the recommendation for each event bases on event similarity from Model 2.

With the help of above content-based recommendation by using events dataset tried to option the recommendations for each user.

user	0	1	2	3	4	5	6	7	8	9
1000293064	3113793753	2030801280	1397608202	2424321825	2027514460	83427780	589257919	1418616930	2915353594	3728974729
1006838695		97217712	1397608202	2006849293	2027514460	589257919	1418616930	2915353594	3728974729	2773204108
1008893291	3113793753	1397608202	2076836172	2027514460	589257919	1418616930	2915353594	3728974729	3104408471	3519129293
1011223331	1397608202	2731002498	2027514460	589257919	1418616930	3728974729	1903444229	2773204108	1279926604	221202458
1016040879	1927168893	3677554347	3562955208	2571012976	835360674	2021591040	4276436069	4040181802	3900421971	1589462668
...	...	...	...	...	...	...	...	...	...	...
988213942	2975196902	97217712	1397608202	2340432094	3728974729	610367576	1638236770	328882352	2780066048	2742780288
990116823	2271507036	97217712	1397608202	2424321825	3428913660	2340432094	3728974729	3104408471	1220666503	1972417022
990496397	97217712	1397608202	2006849293	2027514460	589257919	1418616930	2915353594	3728974729	2773204108	642644408
995603012	3166241555	44776435	3711909804	2126750158	3124812840	3425996117	31247346	1366554501	244219005	46300535
996222490	1888241344	3154390339	3284103018	2232544430	3678607925	1076499062	2571012976	546889678	297856962	92245848

2034 rows × 100 columns

The above shows the events recommended for each user.

Approach Followed for alternate model:

We tried to collect all the event that user has responded interesting. And for all those events with the help of content-based recommendation collected all the list of events that has similarity greater than 0.5.

And limited them to 100 events for each user and suggested that events.

Discussion:

The core components of the system, Model 1 and Model 2, demonstrated efficacy in identifying similar users and events, respectively. Though Model 1 identified users with comparable preferences, allowing for the generation of relevant event recommendations based on similar user behaviour, because it was trained on a sparse dataset, the KNN's performance is a fallacy and can't be utilized to find the user similarity accurately. However, Model 2 excelled in identifying events with similar characteristics based on their cosine similarity, enabling the system to recommend events based on their intrinsic qualities. Thus, for this reason, we have used only the Model 2 for even the recommendation.

Through meticulous data preprocessing and feature engineering, the system optimized the datasets for analysis, resulting in accurate and relevant recommendations. Key features such as event attributes were carefully considered and incorporated into the recommendation models, enhancing recommendation accuracy and relevance.

The approach employed by the system, which is the content-based filtering methodology, proved to be particularly effective in delivering diverse and relevant recommendations. By integrating user behaviour with event characteristics, the system was able to provide users with a comprehensive selection of events tailored to their preferences. The alternate model content-based recommendation is highly accurate than the hybrid model we built earlier.

Looking ahead, ongoing efforts are focused on further refining and optimizing the Event Recommendation System to enhance its accuracy, relevance, and usability. In terms of refining, the Model 1 can be trained with a better dataset, which would allow us to build a better Collaborative model for event recommendation. Strategies such as continual experimentation, and parameter optimization can be employed to elevate the system's performance and ensure that users receive compelling and relevant event recommendations.

The Event Recommendation System represents a valuable tool for empowering users to discover and engage with a diverse array of events effortlessly. By leveraging advanced technology and data-driven insights, the system opens up new avenues for event recommendation, fostering greater engagement, connection, and enjoyment for users worldwide.

5. CONCLUSION

This project target was to achieve an event recommendation using machine learning and data mining. It aims to address the issues facing other recommendations of relying on too basic information without understanding why. The solution that's employed in the event recommendation is that it's a hybrid model that consist of collaborative filtering and content-based filtering. This should be noted that one of the datasets was not appropriate to be used due to a small feature set. This was employed due to the nature of how the datasets is made up of an event dataset that has values of c for their characteristics and a user dataset that consist of a userbase that marks events based on if they're interested in it or not. The data was cleaned by removing was irrelevant such as user's country or the event's latitude and longitude. Thus, the data is focused for explicitly on events and user's interest on the event or related events. With data cleaned and a hybrid model planned and developed, the results were found and noticed a few key points. Firstly, the KNN graphs showed us that that the optimal K value to have the lowest RMSE value would be somewhere between 11 and 21. Compared to existing solutions, the project's solution has good accuracy that works with users reflected in mind. In summary, the project consisted of two model for two different dataset that seen solid results but in the further refining process an alternate model was built that is more accurate than hybrid model.

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